UNIT 8

# INTRODUCTION TO DYNAMICS

## Learning objectives:

- Study of forces and their effect on motion, like acceleration and safety devices.
- Differentiate between Mass and Weight.
- Learn the concepts of acceleration and force, and the Equations of motion.

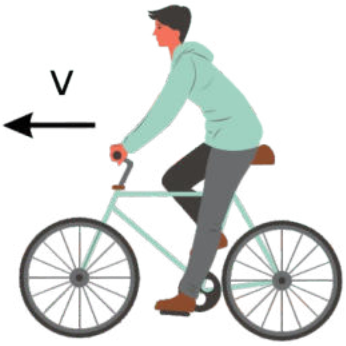
## 8.1 Introduction to Dynamics

### What is Dynamics?

Dynamics studies forces and their effect on motion.

### Importance in Everyday Life

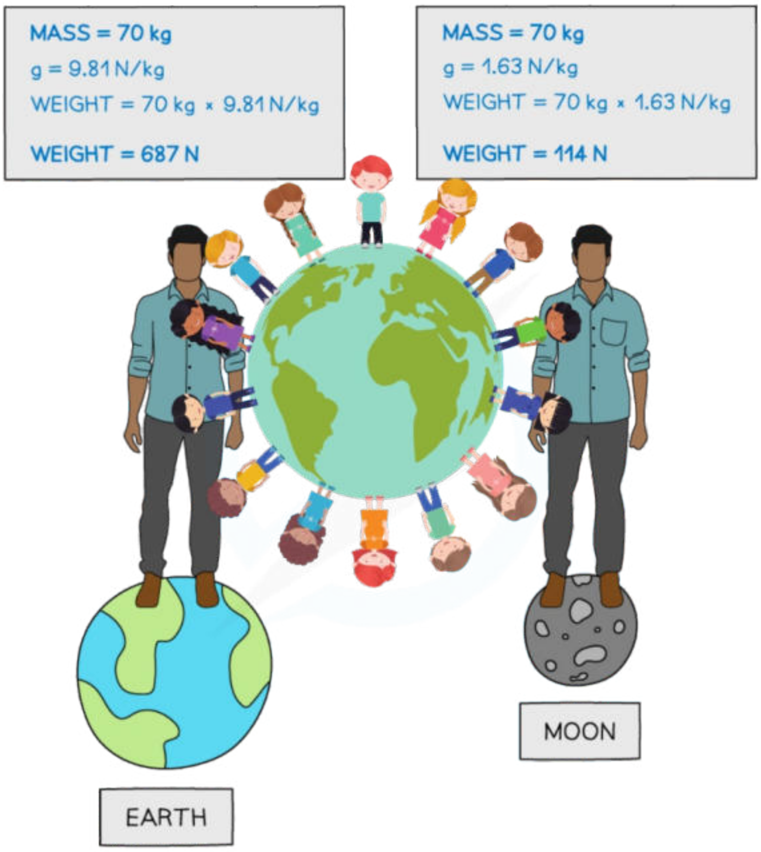
It explains everyday phenomena like acceleration and stopping, and is key in designing machines and safety devices.




## 8.2 Mass and Weight

**Differences between Mass and Weight:**

| Property | Mass | Weight |
|---|---|---|
| Definition | The amount of matter in an object. | The force due to gravity. |
| Unit | Kilograms (kg) | Newtons (N) |
| Changes? | No | Yes (varies with location) |
| Example | A book has a mass of 2 kg. | The book weighs 20 N on Earth. |




## 8.3 Acceleration

### What is Acceleration?

Acceleration is the rate at which an object's velocity changes over time. It occurs when an object speeds up, slows down, or changes direction.

### Formula for Acceleration:

$$\text{Acceleration} = \frac{\text{change in velocity}}{\text{time taken}}$$

$$\text{Acceleration} = \frac{\text{final velocity - initial velocity}}{\text{time taken}}$$

$$\mathbf{a} = \frac{\mathbf{v_f - v_i}}{t}$$

### Where:

- vf = is the final velocity
- vi = is the initial velocity
- t = is the time taken

**Example:**

If a car speeds up from 10 m/s to 20 m/s in 5 seconds, the acceleration is:

$$(a) = \frac{(vf - vi)}{(t)}$$

$$(a) = \frac{(20 - 10)}{(5)}$$

$$a = \frac{\cancel{10}^{2}}{\cancel{5}^{1}}$$

$$a = 2 m/s^2$$

## Types of Acceleration

### 1. Positive Acceleration:

When an object's velocity increases over time.

### 2. Negative Acceleration (Deceleration):

When an object's velocity decreases over time.

### 3. Uniform Acceleration:

When an object's velocity changes by the same amount in equal time intervals.

### 4. Non-Uniform Acceleration:

When an object's velocity changes by varying amounts in equal time intervals.

## 8.4 Equations of Motion

### First Equation of Motion

**Formula**

$$v_f = v_i + at$$

**Derivation**

**Definition of Acceleration:**

Acceleration a is the rate of change of velocity:

$$a = \frac{\text{Change in Velocity } (v_f - v_i)}{\text{Change in Time taken } (t)}$$

**Where:**

**vf =** final velocity,

**vi =** initial velocity,

**t =** time taken.

Rearranging the formula to solve for vf:

$\mathbf{v_f = v_i + at}$

### Second Equation of Motion:

$$S = v_i t + \frac{1}{2} at^2$$

**1. Average Velocity**

$$V_{avg} = \frac{(v_i + v_f)}{2}$$

**2. Displacement**

$$S = V_{avg} \times t = \frac{(v_i + v_f)}{2} t$$

**3. Substitute $v_f$ with $v_i + at$:**

$$S = v_i + \frac{(v_i + at)}{2}$$

$$xt = v_i t + \frac{1}{2} at^2$$

## Third Equation of Motion:

$$V_f^2 = v_i^2 + 2as$$

## 8.5 Force

### What is Force?

A force is a push or pull that can change the state of motion of an object. It has both magnitude and direction.

### SI Unit of Force:

The SI unit of force is Newton (N).

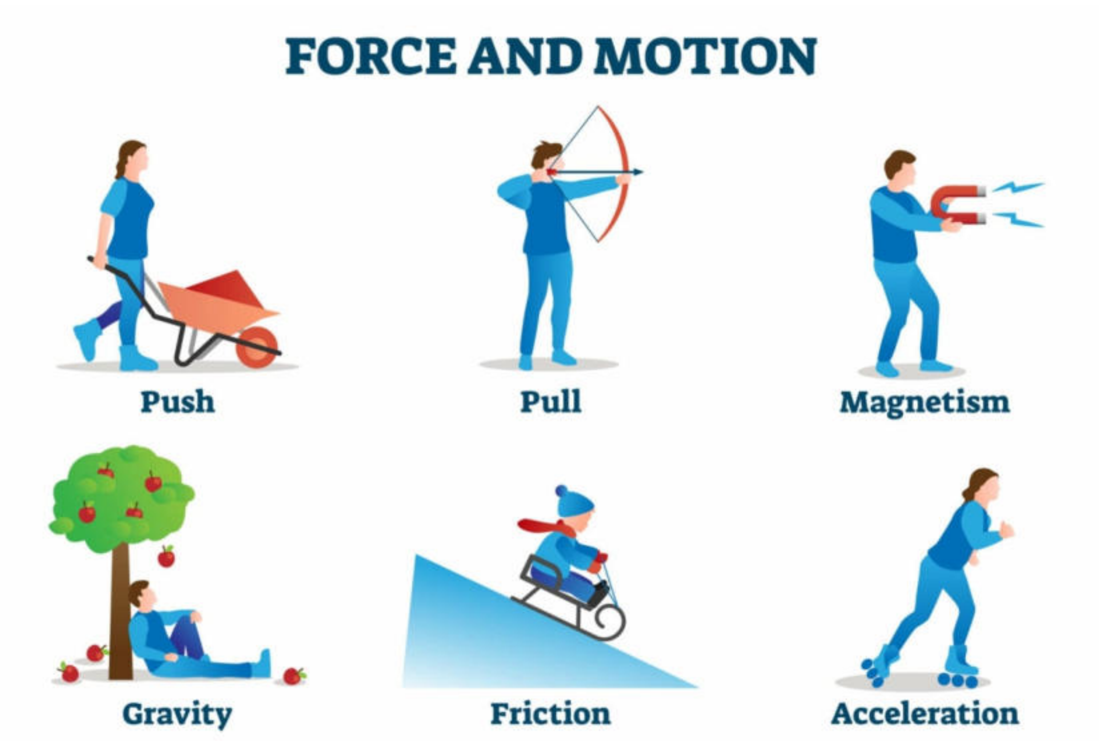
### Types of Forces:

- **Push and Pull:** Forces that move objects.

**Example:** Pushing a door, pulling a rope.

- **Frictional Force:** The force that opposes motion.

**Example:** Friction slows down a sliding book.

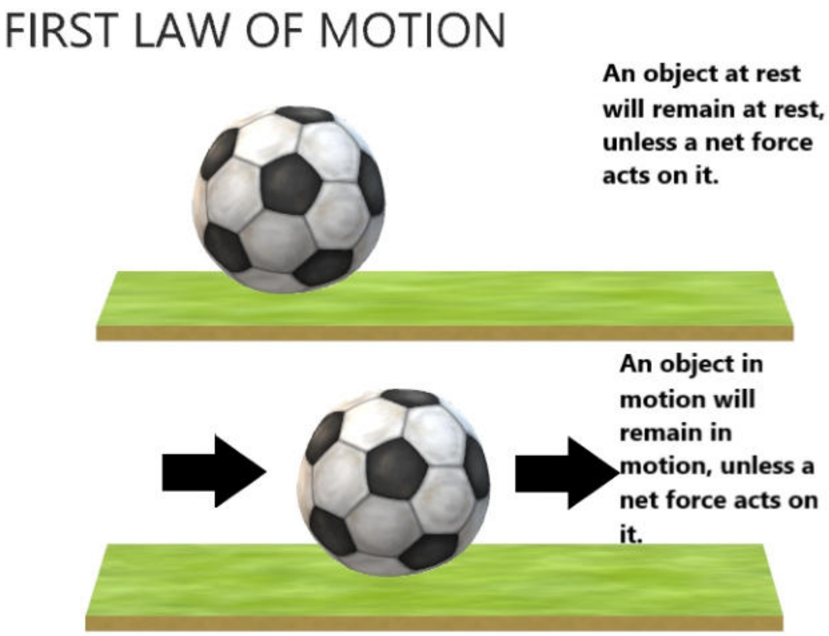


## 8.6 Newton's Laws of Motion

### First Law of Motion (Law of Inertia)

An object remains at rest or in uniform motion unless acted upon by an external force.

**Example:** When a bus stops suddenly, your body continues moving forward (inertia).




### Second Law of Motion

The force acting on an object is equal to its mass multiplied by its acceleration:

$$F=ma$$

**Example:** A heavier car needs more force to accelerate than a bicycle.

**Third Law of Motion**

For every action, there is an equal and opposite reaction. **Example:** A rocket moves upward due to the action of exhaust gases being pushed down.

## 8.7 Friction

### What is Friction?

Friction is the force that resists motion between two surfaces in contact.

### Types of Friction:

• **Static Friction:** Prevents motion between stationary objects.

**Example: A book resting on a table.**

• **Sliding Friction:** Opposes motion when an object slides over a surface.

**Example:** A book sliding on a table.

• **Rolling Friction:** The friction when an object rolls. Example: A ball rolling on the ground.

### Advantages of Friction:

• Prevents slipping while walking.

• Enables safe braking in vehicles.

### Disadvantages of Friction:

• Causes wear and tear on objects.

• Reduces machine efficiency.

### Ways to Reduce Friction:

• Lubrication (e.g., oil in engines).

• Smoother materials.

## 8.8 Applications of Dynamics

**Safety Devices:**

• Airbags and seatbelts use Newton's laws to protect us during accidents by reducing the impact force.

**Uses of Friction:**

• **Walking:** Friction between shoes and the ground prevents slipping.

• **Vehicle Brakes:** Friction helps stop a vehicle.

# EXERCISES:

## Fill in the blanks.

**1.** The SI unit of force is ______________.

**2.** ______________ is the force that opposes motion.

**3.** The formula for acceleration is ______________.

**4.** The unit of mass is ______________.

**5.** The unit of weight is ______________.

**6.** An object's acceleration depends on ______________ and ______________.

**7.** Newton's second law is expressed as ______________.

## Tick(✓) for True and cross (x) for False statements.

**1.** Friction is the force that helps us walk.

**2.** The unit of force is kilogram.

**3.** Acceleration is the rate of change of velocity.

**4.** A car moving at a constant speed has zero acceleration.

**5.** Newton's third law states that every action has an equal and opposite reaction.

**6.** Mass is a measure of the amount of matter in an object.

**7.** Gravitational force pulls objects away from the Earth.

**8.** Inertia is the tendency of an object to resist changes in its motion.

## Choose the correct answer.

**1.What is the SI unit of force?**

a) Newton b) Kilogram

c) Metre d) Second

**2.Which of the following is an example of rolling friction?**

a) A book sliding on a table

b) A ball rolling on the ground

c) A car moving on a road

d) A box resting on the floor

**3.What does the formula F=ma represent?**

a) First law of motion b) Second law of motion

c) Third law of motion d) Law of inertia

**4.Which of the following is true for acceleration?**

a) It only occurs when an object speeds up.

b) It is the rate of change of velocity.

c) It can only be negative.

d) It is independent of time.

## Answer the Following Questions:

### Short-Answer Questions.

1. What is dynamics?
2. What is force? Write the unit and its types with examples.
3. Write the difference between mass and weight.
4. How do seatbelts and air bags work according to Newton's laws of motion?

### Long-Answer Questions.

1. What is acceleration? Write its formula, unit, example and types.
2. What is friction? Write the types, advantages and disadvantages of friction.
3. Write the statements of Newton's laws with examples.

## Newton's law activity

### ACTIVITY

- **Inertia Test:** Place a coin on a card. Flick the card out of the way and observe what happens to the coin (shows inertia).
- **Project:** Conduct an experiment with rolling objects on ramps of different materials (wood, plastic, metal). Measure how far they travel and discuss

### Critical Thinking

Why is friction important for walking and driving but a disadvantage in machines? How can engineers reduce friction in engines and other machines?

**Numericals:**

- A car starts from rest and accelerates at a rate of $3 m/s^2$ for 5 seconds. Find its final velocity.
- A truck is moving at an initial velocity of 10m/s and accelerates at $1.5 m/s^2$ for 4 seconds. What is its final velocity?
- A cyclist increases their velocity from 5 m/s to 25 m/s in 10 seconds. Calculate the acceleration.
- A car starts from rest and accelerates at $24 m/s^2$ for 6 seconds. What is its final velocity?

## Types of Friction

**Directions:** Write definition for friction. Complete the Examples column in the table below.

**Friction is** ____________________________________________

| Examples | Type of Friction | Characteristics |
|---|---|---|
| | **Static** | Frictional Force causing object(s) to stay at rest. No Movement |
| | **Sliding** | Frictional Force between two objects causing the object(s) to slide past each other. |
| | **Fluid** | Frictional Force between an object and a fluid. |
| | **Rolling** | Frictional Force between two objects causing the object(s) to roll. |